Linear Regression:

Automobile Fuel Efficiency Prediction

The automobile industry wants to develop a predictive model to estimate the fuel efficiency of different car models based on their engine and vehicle specifications.

You are provided with the Auto MPG Dataset, which contains information about various automobiles, including engine displacement, horsepower, vehicle weight, acceleration, and model year.

Your task is to build a Linear Regression model from scratch using NumPy to predict the miles per gallon (mpg) of automobiles and visualize the regression results using Matplotlib.

Dataset Details:

Dataset size: 398 samples with 8 attributes.

Column	Description
mpg	Miles per gallon (target variable)
cylinders	Number of engine cylinders
displacement	Engine displacement in cubic inches
horsepower	Engine horsepower
weight	Vehicle weight in pounds
acceleration	Time taken to accelerate, measured in seconds
model year	Model year of the automobile
origin	Region where the car was manufactured
car name	Name of the automobile

What to Do?

• Implement Linear Regression from scratch using NumPy. (Do not use sklearn)

• Load and prepare the dataset Using Pandas.

• Train your Linear Regression model and visualize the regression fit or residuals using Matplotlib.

• Output the final performance score you obtain.

Logistic Regression:

The National Socioeconomic Research Institute (NSRI) aims to develop a predictive model to determine whether an individual earns **more than $50,000** per year based on their demographic and employment attributes.

You are provided with the dataset adult.csv, which contains census data including age, education, occupation, and other socioeconomic indicators.

Your task is to implement Logistic Regression from scratch using NumPy to predict whether a person earns >50K or <=50K annually. You will preprocess the data, train your model, visualize results, and evaluate its performance.

Column	Description
age	Age of the individual
workclass	Type of employment (e.g., Private, Self-emp, Govt)
fnlwgt	Final weight assigned to the record
education	Highest level of education attained
education-num	Numeric representation of education level
marital-status	Marital status of the individual
occupation	Type of occupation
relationship	Relationship status (Husband, Wife, etc.)
race	Race of the individual
sex	Gender of the individual
capital-gain	Capital gains from investments
capital-loss	Capital losses from investments
hours-per-week	Average working hours per week
native-country	Country of origin
income	Target variable (>50K = 1, <=50K = 0)

What to Do

• Implement Logistic Regression from scratch using NumPy (do not use sklearn or other ML libraries).

• Start by loading and preprocessing the dataset Using Pandas.

• Train your Logistic Regression model and visualize important aspects (such as loss over iterations or feature relationships) using Matplotlib.

• Output the accuracy your model achieves on the test set.